PTX130W

Wireless Charging Poller IC

The PTX130W is a powerful and efficient NFC transmitter system-on-chip for NFC wireless charging applications alongside data communication. PTX130W features improved key performance parameters such as output power and RX sensitivity (-80dBc[1]). While eliminating EMC filter and matching components the PTX130W enables simple integration and compact design without the complexity associated with existing solutions (dual resonating circuits composed by EMC filter and antenna). The superior RF performance of PTX130W enables small antenna design, fast charging and allows flexible placement of Poller and Listener antennas.

Features

- The architecture enables high and efficient power transmission with accurate digital shaping programmability
- Architecture enabling up to 18% efficiency improvement compared to previous generation (PTX100W)
- Harvesting up to 1W on the NFC WLC Listener
- No EMC filter required due to sinewave output driver and Direct Antenna Connection (DiRAC)
- -80dBc RX sensitivity with full dynamic range due to DiRAC
- Automated Power Control Loop for system efficiency optimization
- WLC Poller functionalities
- WLC Poller with extended functionality[2]
- Impedance change detection
- On-Chip processing of time critical commands
- Output power regulation with 1% step

PTX130W NFC Charging IC enables key improvements in customer care-about such as:

Charging performance:

Groundbreaking architecture enabling efficient power transmission due to direct antenna connection enables fast charging with small antenna design.

Poller Listener antenna misplacement:

Power harvested constant even over volume due to constant loading effect (EMC filter removal).

Enables flexible placement of Poller and Listener antennas in x, y and z axis.

Manufacturability:

- No need for bulky and performance-limiting external components of the EMC filter (no tolerances issue introduced) minimizing the performance differences between final devices.
- Reduced number of matching components allow a lower antenna matching impedance, resulting in higher output power.
- Accurate gain receiver and power steps control due to digital architecture enabling tighter production control to give more margin for new use cases.

Applications

PTX130W is optimized for applications such as NFC Wireless charger devices for Wearables, Smart Watches, Smart Glasses, Hearing Aids, Smart Rings, etc

[1] Receiver input dynamic range and sensitivity are defined in Table 5.

[2] Such as proprietary commands for transparent data channel

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Contents

1 Block Diagram ... 3

2 Pinning Information ... 4

2.1 Pin Diagram ... 4

2.2 Pin Description ... 4

3 Electrical Characteristics ... 7

3.1 Absolute Maximum Ratings ... 7

3.2 Electrical Characteristics ... 7

4 Functional Description ... 8

4.1 Protocol ... 8

4.1.1 Foreign object detection (FOD) ... 9

4.2 Power Management ... 9

4.3 Power Supply Concept ... 9

4.3.2 SEN-pin ... 10

4.3.3 Supply Ramp-Up Sequence ... 10

4.3.4 Energy states ... 11

4.4 Clock Concept ... 12

4.4.1 Low Power Oscillator (LPO) ... 12

4.4.2 External Reference Clock ... 12

4.4.3 Crystal Oscillator (XTAL) ... 12

4.4.4 Phase Locked Loop (PLL) ... 12

4.5 Contactless Interface ... 12

4.5.1 Analog/Digital Receiver ... 13

4.5.2 Analog/Digital Transmitter ... 13

4.5.3 Low Power Card Detection ... 13

4.6 Auxiliary Features ... 14

4.6.1 GPIO functionality ... 14

4.6.2 Auxiliary Digital-to-Analog Converter (AUX-DAC) ... 14

4.6.3 Temperature Sensor and Over-Temperature Protection ... 14

4.7 Programmable Control Logic ... 14

4.7.1 Hardware Accelerator ... 14

4.7.2 PTX130W device control firmware ... 14

4.8 SPI Interface ... 14

5 FW/SW Functionality ... 16

5.1 NSC Interface ... 16

5.2 SW Split Stack ... 17

6 Reference Schematic ... 18

7 Package Information ... 19

7.1 Package Marking ... 19

7.2 Package Drawing and Dimension ... 20

8 Revision History ... 20

1 Block Diagram

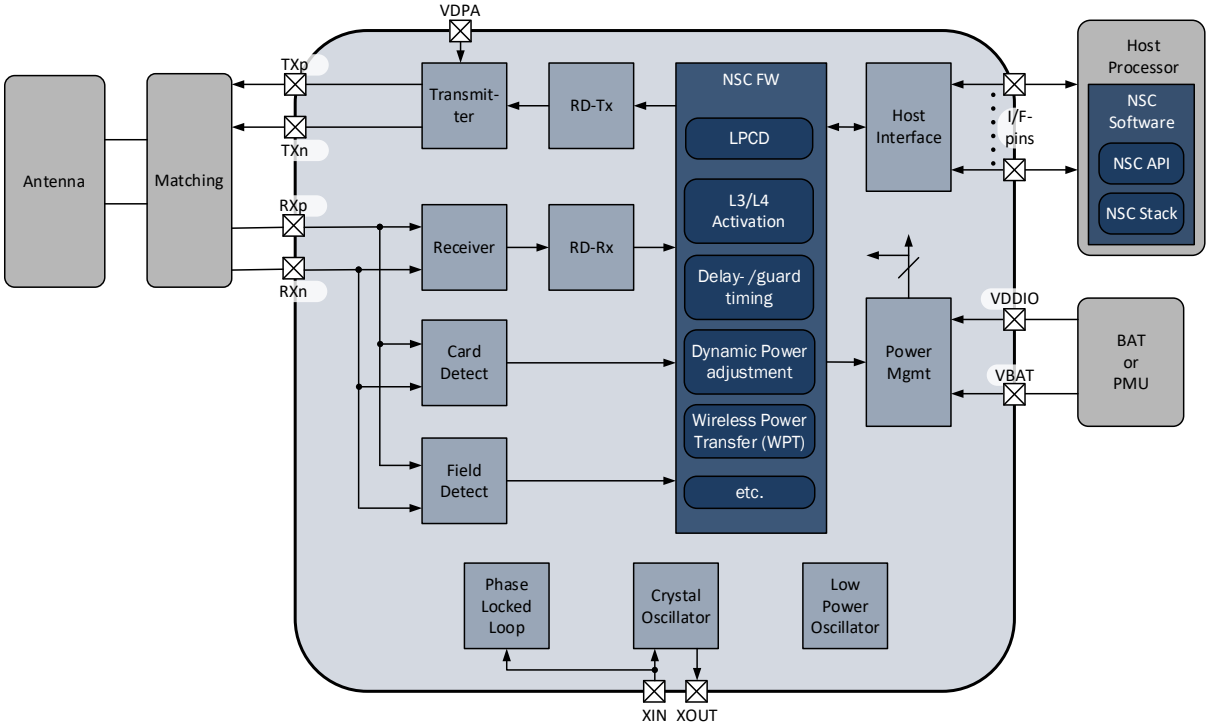


Figure 1: Block Diagram

2 Pinning Information

2.1 Pin Diagram

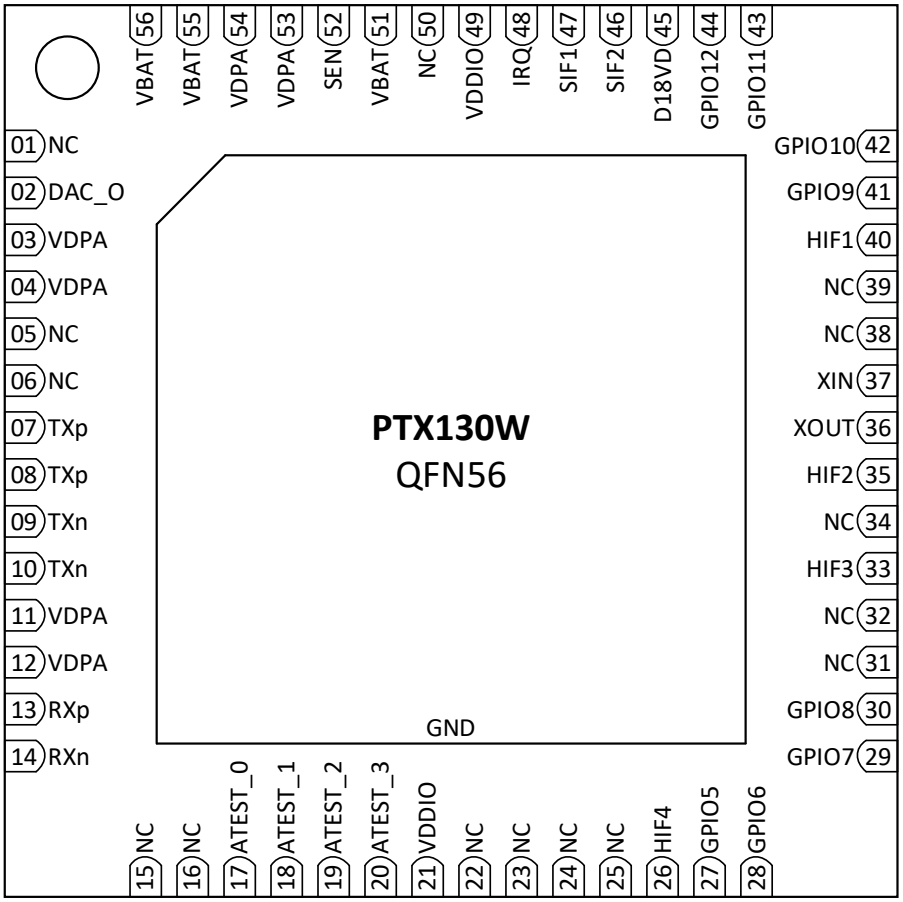


Figure 2: Pin diagram (transparent top view)

2.2 Pin Description

Pin number	Pin name	Signal type	Description
1	NC	-	Not connected, leave open
2	DAC_O	Analog out	AUX-DAC output voltage
3	VDPA	Supply	Transmitter block supply
4	VDPA	Supply	Transmitter block supply
5	NC	-	Not connected, leave open
6	NC	-	Not connected, leave open
7	TXp	Analog out	Transmitter output pin p
8	TXp	Analog out	Transmitter output pin p
9	TXn	Analog out	Transmitter output pin n
10	TXn	Analog out	Transmitter output pin n
11	VDPA	Supply	Transmitter block supply

12	VDPA	Supply	Transmitter block supply
13	RXp	Analog in	Receiver block input pin p
14	RXn	Analog in	Receiver block input pin n
15	NC	-	Not connected, leave open
16	NC	-	Not connected, leave open
17	ATEST_0	Analog out	Analog test mux
18	ATEST_1	Analog out	Analog test mux
19	ATEST_2	Analog out	Analog test mux
20	ATEST_3	Analog out	Analog test mux
21	VDDIO	Supply	Pad supply
22	NC	-	Not connected, leave open
23	NC	-	Not connected, leave open
24	NC	-	Not connected, leave open
25	NC	-	Not connected, leave open
26	HIF4	Digital in/out	SPI: MISO
27	GPIO5	Digital in/out	General purpose digital IO pin
28	GPIO6	Digital in/out	General purpose digital IO pin
29	GPIO7	Digital in/out	General purpose digital IO pin
30	GPIO8	Digital in/out	General purpose digital IO pin
31	NC	-	Not connected, leave open
32	NC	-	Not connected, leave open
33	HIF3	Digital in/out	SPI: MOSI
34	NC	-	Not connected, leave open
35	HIF2	Digital in/out	SPI: SCK
36	XOUT	Analog out	Xtal oscillator output
37	XIN	Analog in	Xtal oscillator input / Reference clock input
38	NC	-	Not connected, leave open
39	NC	-	Not connected, leave open
40	HIF1	Digital in/out	SPI: NSS
41	GPIO9	Digital in/out	General purpose digital IO pin
42	GPIO10	Digital in/out	General purpose digital IO pin
43	GPIO11	Digital in/out	General purpose digital IO pin
44	GPIO12	Digital in/out	General purpose digital IO pin
45	D18VD	Supply	Decoupling of core supply
46	SIF2	Digital in	Select interface type bit 2 – Shall be set to GND for SPI
47	SIF1	Digital in	Select interface type bit 1 – Shall be set to GND for SPI
48	IRQ	Digital out	Interrupt request to host
49	VDDIO	Supply	Pad supply

50	NC	-	Not connected, leave open
51	VBAT	Supply	Main supply
52	SEN	Analog in	System enable input pin
53	VDPA	Supply	Transmitter block supply
54	VDPA	Supply	Transmitter block supply
55	VBAT	Supply	Internal functional blocks supply
56	VBAT	Supply	Internal functional blocks supply
	GND	Supply	The exposed pad at the back of QFN is used as GND

Table 1: Pin description

3 Electrical Characteristics

3.1 Absolute Maximum Ratings

Symbol	Parameter	Conditions	Min	Typ	Max	Unit	Remarks
V_{SMAX}	VBAT and VDPA supply pins		-0.5		5.5	V	
T_J	Junction temperature		-40		125	°C	
T_S	Storage temperature		-40		150	°C	
$R_{th(ja)}$	Thermal resistance junction to air				35	K/W	Based on JESD-51
P_{TOT}	Total power dissipation on chip				1.5	W	
$V_{ESD(HBM)}$	electrostatic discharge voltage; Human Body Model (HBM)	1500 Ohm, 100 pF; JESD22-A114-B			1000	V	All pins except RXp/RXn
$V_{ESD(HBM),RX}$	electrostatic discharge voltage; Human Body Model (HBM)	1500 Ohm, 100 pF; JESD22-A114-B			750	V	RXp/RXn
$V_{ESD(CDM)}$	electrostatic discharge voltage (Charge Device model)	Field induced model; EIA/JESC22-C101-C			750	V	
I_{LU}	Latch up	AEC-Q100 (transient current)	100			mA	
V_{INMAX}	Maximum input voltage on digital pins		-0.3		V_{DDIO} +0.3	V	

Table 2: Absolute Maximum Ratings

3.2 Electrical Characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit	Remarks
T_A	Ambient temperature		-40		70	°C	
T_J	Junction temperature		-40		120	°C	
V_{BAT}	Supply voltage		2.7		5.5	V	
V_{DDIO}	Pad supply voltage		1.62		5.5	V	
V_{VDPA}	Transmitter supply		2.5		V_{BAT}	V	
V_{SEN_H}	System Enable (SEN) pin high-level voltage range		1.62		V_{BAT}	V	
V_{IH}	GPIO pins high level input voltage		0.75* V_{DDIO}		V_{DDIO}	V	
V_{IL}	GPIO pins low level input voltage		0		0.15* V_{DDIO}	V	
V_{OH}	GPIO pins high level output voltage[3]		V_{DDIO} -0.5		V_{DDIO}	V	
V_{OL}	GPIO pins low level output voltage		0		0.45	V	

[3] *VOH measurement condition: Iload=2.8mA with VDDIO=4.5V; Iload=0.4mA with VDDIO=1.62V.*
VOL measurement condition: Iload=2.1mA with VDDIO=4.5V; Iload=0.6mA with VDDIO=1.62V.

Table 3: Operating Range

Symbol	Parameter	Conditions	Min	Typ	Max	Unit	Remarks
I_{PD}	Power down current consumption	V_{BAT} = 3.6V SEN pin = '0'		3		uA	
I_{STBY}	Standby current consumption	V_{BAT} = 3.6V		15		uA	
I_{CD_rd}	Low power card detection current consumption	V_{BAT} = 5.5V		100		uA	2Hz polling frequency with optimized reading distance
I_{VDPA}	Transmitter supply input current				650	mA	

Table 4: DC characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit	Remarks
$V_{in_rx_rd}$	RX carrier signal reader		100 m		50	V_{P-P}	Differential
$S_{in_RX_rd}$	RX sensitivity AM			-80[4]		dBc	$V_{in_rx_rd}$ ≥25Vpp
$H_{out_tx_rd}$	Transmitter output harmonics			-60		dBc	
C_{out_tx}	Transmitter serial output capacitance		418	440	462	pF	

Table 5: Receiver / Transmitter Characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit	Remarks
f_{ref_clk}	Reference clock input frequency			27.12		MHz	
$V_{ref_clk_low}$	Reference clock input voltage low		0		400	mV	
$V_{ref_clk_high}$	Reference clock input voltage high		1.4		1.95	V	
Δf_{ref_clk}	Reference clock frequency tolerance		-50		+50	ppm	Tested for 27.12MHz
$DC_f_{ref_clk}$	Reference clock duty cycle		40	50	60	%	

Table 6: Reference Input Frequency Requirements

4 Functional Description

4.1 Protocol

The PTX130W together with the SDK package implements Wireless Charging functionality.

The PTX130W uses standard NFC communication to exchange information with the WLC Listener device. Prior to starting data exchange with the WLC Listener device, the standard NFC Link Establishment (NFC LE) procedure

[4] To achieve -80dBc RX sensitivity, VBAT shall not have voltage ripple higher than 150mVrms (VBAT = 5V) around the carrier frequency (±1MHz)

is executed. The procedure involves NFC-polling based on LPCD (Low Power Card Detection), protocol-based device discovery and device activation.

After link establishment, the WLC Poller (PTX130W) executes Wireless Charging Control Activation, in which the WLC Poller identifies WLC Listener devices, and in the next step WLC Listener capabilities are read. The following phases are Wireless Charging Control (WCC) and Wireless Power Transfer (WPT). In Wireless Charging Control the WLC Listener negotiates with the WLC Poller for the power level and request for the next Wireless Power Transfer (WPT) phase. The WLC Poller information is provided to the listener via the WLCP_INFO NDEF message and the WLCL_CTL NDEF is used for WLC Listener controlling the WLC Poller.

Bytes from the WLC Poll Information Record of the PTX130W are either SDK configured or dynamic (e.g. P_TX and CUR_POWER_STEP).

4.1.1 Foreign object detection (FOD)

The PTX130W supports NFC Polling FOD, which detects foreign objects that can communicate as NFC Forum devices in listen mode. The NFC Polling FOD is part of the link establishment phase, and it is enabled via the provided SDK.

The PTX130W also supports Background FOD based on impedance change detection. The Background FOD provides a mechanism to detect foreign objects that are inserted into the wireless charging operating volume (WLC OV) during wireless power transfer phase (WPT).

4.2 Power Management

The power management unit is the central circuit of the PTX130W responsible for providing all necessary reference voltages and currents, generating the internal supply domains, implementing the power-up sequence, and controlling the transitions between the different energy states.

4.3 Power Supply Concept

PTX130W has 4 externally accessible supply domains which are described as follows:

4.3.1.1 VBAT

Main supply domain from which all functional blocks except the transmitter are supplied. To operate the IC this supply must always be present.

4.3.1.2 D18VD

The internally generated core supply is routed to pin D18VD for external supply decoupling. Hence external supplies are not allowed to be connected. For recommended decoupling components refer to reference schematic in Section 6.

4.3.1.3 VDDIO

Pad supply for all GPIO- and HIF/SIF-pins. It must be present during start-up for proper host-interface selection and afterward for host interface communication. In power-down mode, the voltage may be removed from this domain, but the pin shall not be pulled to VS (i.e. to be put to HiZ).

If for power saving reasons at any point in time the VBAT domain is switched off, then V_{DDIO} must also be removed.

4.3.1.4 VDPA

The VDPA net is the transmitter supply. It can be connected on the same supply as VBAT.

Note: VDPA supply voltage shall be equal or lower than VBAT.

4.3.2 SEN-pin

The SEN-pin is used to boot up the PTX130W (logic-high level) or to bring the IC into power-down state (logic-low level). Detailed ranges for the logic-levels are given in Table 3.

Note that SEN-input voltage must never exceed V_{BAT}. For safe operation, SEN shall change to a logic low if V_{BAT} drops below V_{BATmin} (value defined in Table 3).

Logic-low level pulses with smaller than 3.4 us pulse width on SEN-pin will not change the current energy state of the IC. Negative pulses that are slightly longer will first reset the internal state and for even longer negative pulses the IC changes the energy state to power-down mode. The exact pulse width below which a reset is triggered, and above which power-down mode is entered depends on the blocking capacitor value on D18VD-pin.

4.3.3 Supply Ramp-Up Sequence

For the supply ramp-up two sequences are proposed: a default sequence with SEN pin at logic-low for relaxed timing constraints between V_{BAT} and V_{DDIO}, and a sequence with SEN pin connected to V_{BAT} for simpler configuration and faster start-up (VVDPA can be ramped up same time as VBAT or later).

Ramp-up sequence with SEN at logic-low (transition into power-down mode):

- First, the battery supply V_{BAT} shall be ramped up – the IC remains in power-down mode
- V_{DDIO} shall be ramped up. V_{DDIO} may ramp concurrently to V_{BAT}.
- V_{DDIO} supply and a stable state of SIF1/SIF2 pins shall latest be available when the SEN pin voltage reaches a valid logic-high level. These timings also apply when leaving power-down mode with disabled V_{DDIO}.

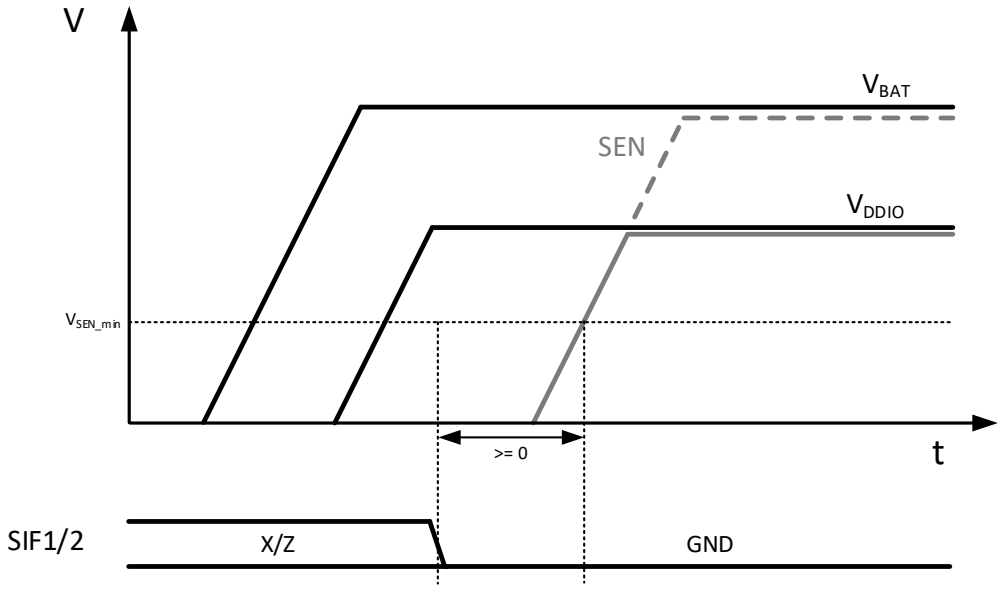


Figure 3: Supply Ramping, SEN at logic-low

Ramp-up sequence with SEN at logic-high (transition into full-power mode):

- First, the battery supply V_{BAT} together with SEN pin voltage shall be ramped up.
- V_{DDIO} shall be ramped and shall be at V_{DDIO_min} latest 100 us after V_{BAT} exceeds V_{BATmin} (i.e. 2.7 V). Latest at this point also the state of the SIF1/SIF2 pins shall be stable.

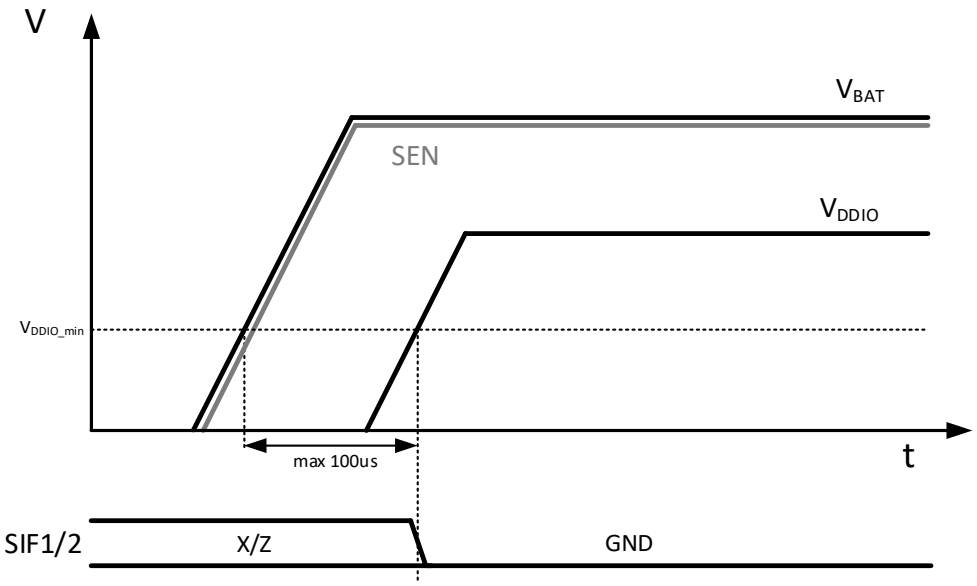


Figure 4: Supply Ramping, SEN==VBAT

In typically applications, after SEN, V_{DDIO} and SIF pins reach their stable level, PTX130W is ready for operation in less than 1ms.

4.3.4 Energy states

To support the implementation of flexible system power consumption profiles, PTX130W offers different energy states, each being a trade-off between functionality and power consumption.

4.3.4.1 Full-Power Mode

This is the main operating mode of PTX130W, in which all internal supply domains are ramped up and all internal clocks are running. This mode is activated by applying a logic-high level at SEN-pin.

In this mode, PTX130W is fully active and can communicate with the host controller via the host interface.

4.3.4.2 Power-Down (PD) Mode

Power down mode is enabled by applying a logic-low level at SEN-pin. In this mode, the PTX130W consumes its lowest power and does not react to any external events. All GPIO-pins and the SIF1/SIF2-pins are switched to HiZ-state.

After leaving this mode the internal state of PTX130W is reset.

4.3.4.3 Standby (STBY) Mode

The standby mode of PTX130W enables low power consumption with the possibility to wake up in response to selected external events such as Low Power Card Detection or Host-interface activities. In this mode, the internal state is fully maintained.

This mode is only available when there is a logic-high level on SEN-pin, and it is autonomously enabled by the device depending on configuration settings.

Events triggering the wakeup of the IC from standby mode include:

- Activity on Host-interface
- Execution of Polling/LPCD procedure

4.4 Clock Concept

In PTX130W a low-power oscillator (LPO), a crystal oscillator (XO) and a phase-locked loop (PLL) are the main blocks responsible for generating the necessary internal clocks in the various modes.

The reference clock for PTX130W can either be provided from an external clock source or a crystal oscillator. Out of this clock, the PLL subsequently derives the system frequency of 13.56 MHz.

4.4.1 Low Power Oscillator (LPO)

The low-power oscillator is the lifeline of PTX130W, and its 125 kHz-clock is particularly employed during IC ramp-up and in standby mode.

4.4.2 External Reference Clock

An external reference clock shall be applied at pin XIN and frequency of 27.12 MHz is supported. Note, PTX130W will boot without the external clock being available by means of an internal clock.

4.4.3 Crystal Oscillator (XTAL)

The internal crystal oscillator is designed for crystal types with a resonant frequency of 27.12MHz, which shall be externally connected between XIN- and XOUT pins. The oscillator will also work with other crystals around this frequency - please contact Panthronics for support.

The PTX130W features internal caps of 6pF each from XIN- and XOUT-pins to VS to act as crystal load capacitors.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit	Remarks
f_c	Frequency			27.12		MHz	
Δf_c	Frequency accuracy		-50		+50	ppm	
ESR	Equivalent serial resistance				150	Ω	
C_L	Load capacitance			6		pF	

Table 7: Crystal requirements

4.4.4 Phase Locked Loop (PLL)

A fractional-N PLL produces the core clock that is used to derive all the needed internal clocks.

A Delta-Sigma Modulator (DSM) is used to program the N division word of the fractional-N PLL. This allows very fine frequency resolution of all clocks and output frequencies.

4.5 Contactless Interface

Lower-level functionality up to communication framing is available through the Contactless Frontend, higher level functionality is implemented via the RF-Subsystem of the Hardware Accelerator unit (see 4.7.1)

Figure 5 gives a block level overview of the Contactless Interface. At TRXp / TRXn pins the device is connected via a matching network to the antenna of the system.

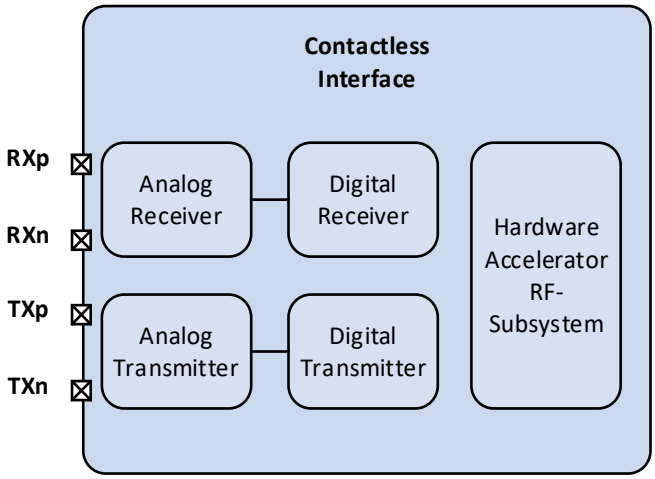


Figure 5: Contactless Interface

4.5.1 Analog/Digital Receiver

The receiver, consisting of an analog and a digital part, is responsible for reception, demodulation and signal processing of incoming commands and data from a PICC. The analog receiver is based on an I/Q-architecture in the RF-domain, which is followed by a baseband chain with programmable gain amplifiers and filters to properly adjust the received signal from varied sized antennas to the full-scale range of the ADC. In the digital receiver, the received information is extracted on a bit and frame-level to finally obtain the transmitted command/data.

4.5.2 Analog/Digital Transmitter

In PCD mode the transmitter is generating the RF-field and consequently amplitude-modulating the PCD commands on the RF-carrier according to the selected communication type.

The analog transmitter itself consists of a digital-like topology, which allows to directly output sinusoidal carriers with high spectral purity and high efficiency. Therefore, EMC-filters, which are required in conventional products in the matching network of Reader-antennas, can be omitted.

The digital transmitter is responsible for encoding the command/data to be sent, applying the respective modulation according to the selected communication type and synchronization of the modulation with the carrier.

The output power of the transmitter can either be varied by the transmitter block by adjusting the sinusoidal amplitude with 1% granularity, or by changing the supply voltage VDPA of the transmitter.

4.5.3 Low Power Card Detection

To optimize the power consumption for low power applications, PTX130W supports Low Power Card Detection (LPCD) feature. LPCD is used to check if a PICC is within the communication range without immediately starting full polling loop. Only when a PICC is present, normal polling will be initiated, otherwise PTX130W goes back to standby. The interval for LPCD is a configurable parameter and can be optimized for different applications.

To check the amplitude and phase of the antenna and determine if a PICC is present, only a reduced set of the hardware blocks are enabled in LPCD mode. Additionally, the active time of Transmitter is kept to a minimum. Compared with normal polling mode which executes communication commands according to standard and requires Transmitter to be active for at least several milliseconds, in LPCD mode the transmitter is only active for less than 100us.

The power consumption in LPCD mode can be further optimized by adjusting the output amplitude and pings interval.

Thanks to the split-stack architecture, the Low Power Card Detection is performed by PTX130W autonomously. Once LPCD mode is configured, the Host MCU is not required anymore, thus can go to lower power mode. In case a PICC is discovered, the host will be notified by PTX130W.

4.6 Auxiliary Features

4.6.1 GPIO functionality

PTX130W offers eight general purpose IOs which can serve either as input or output. Pins are push-pull pads with option to use it as open-drain, programmable internal pull-up, and input Schmitt-Trigger. GPIO functionality is enabled by API functions in the SDK.

4.6.2 Auxiliary Digital-to-Analog Converter (AUX-DAC)

PTX130W comprises a 5-bit general-purpose digital-to-analog converter (DAC), which operates from the VBAT domain and whose output is available at the DAC_O pin.

4.6.3 Temperature Sensor and Over-Temperature Protection

PTX130W features an on-chip temperature sensor that continuously monitors the die temperature. In case the temperature exceeds a configurable threshold, the transmitter is automatically disabled.

4.7 Programmable Control Logic

To fulfil today and future requirements of contactless systems in respect of transaction time, PTX130W provides functions to cover a great variety of higher-level functionalities like ISO-DEP frame de-/composition, automatic frame re-transmission, WTX handling, EMD handling, etc.

These functions are implemented by a programmable control logic circuit – the PTX Hardware Accelerator – guaranteeing great execution speed by keeping flexibility by means of updates/upgrades.

For WLC application and Power Adaptation are also handled by the PTX Hardware Accelerator due to the strict timing requirement.

4.7.1 Hardware Accelerator

The Hardware Accelerator executes the program downloaded to the PTX130W code-memory. It has access to all internal status information as well as the configuration mechanisms.

After downloading the micro-code (uCode) the accelerator can be enabled. Subsequently, it accepts NSC (NFC-Soft-Controller) commands for handling all tasks.

4.7.2 PTX130W device control firmware

4.7.2.1 uCode Download Mechanism

Before enabling the hardware accelerator, the devices uCode must be downloaded. This is done via the host-interface by utilizing the Write-Instruction functionality.

4.7.2.2 Accelerator Enable

Activating the accelerator is achieved by writing the corresponding command to the device.

4.7.2.3 Soft-Reset

PTX130W provides a Soft-Reset functionality that resets all digital blocks and disables all analog blocks (same condition as after boot).

4.8 SPI Interface

The host interface is designed for typical interface supply voltages used by micro-controllers in the range of 1.8 V to 5 V (+/- 10%) which must be supplied by the host via the VDDIO pin.

Every communication is driven by the master (the host). To initiate a transfer from PTX130W to the host, a dedicated IRQ pin is available which the device uses to notify the host that it wants to communicate.

Table 8 specifies the pin assignment for SPI interface, all HIF-interface pins are used during the communication.

Pin name	SPI interface pin function
HIF1	NSS
HIF2	SCK
HIF3	MOSI
HIF4	MISO

Table 8: Pin Assignment for SPI Communication

PTX130W implements a standard SPI interface supporting the SPI mode 0 (CPOL = 0, CPHA = 0) up to 10MHz, i.e. the clock must be low when data changes and data is captured at the leading clock edge after NSS is de-asserted.

It uses 4 signal lines for communication:

- Not-Slave-Select (NSS): Active low input to select the device. Communication is initiated by pulling NSS low. When NSS is high the data output MISO is disabled
- Serial Clock (SCK): Clock input for the SPI interface.
- Master-Out-Slave-In (MOSI): Serial data line from host (master) to PTX130W. Data is registered at the positive edge of the clock.
- Master-In-Slave-Out (MISO): Serial data line from PTX130W to host (master). Data is shifted on negative edge of the clock.
  Note: The pin is permanently driven by strong push-pull driver between logic “0” and “1” states, consequently multidevice sharing the MISO line requires an additional external circuit.

Figure 6 below illustrates a standard SPI transfer with mode 0. The communication starts by pulling NSS line low.

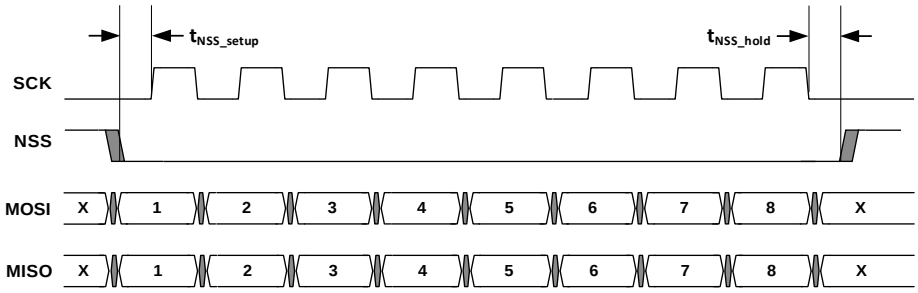


Figure 6: SPI mode 0 (CPOL, CPHA = 0)

Data transfers must always be byte aligned, i.e. the number of bits transmitted is a multiple of 8. Furthermore, the minimum numbers of bytes per frame are 2 (1 header byte + 1 data byte) and frames must be transmitted at once – without pulling NSS high in between.

Further timing characteristics for SPI interface are specified in Table 9.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit	Remarks
f_{SCK}	SPI clock frequency				10	MHz	
DC_{SCK}	SPI clock input duty cycle		40	50	60	%	
t_{NSS_setup}	Setup time from NSS low to rising edge of SCK		25			ns	
t_{NSS_hold}	Hold time from falling edge of SCK to NSS high		25			ns	

Table 9: SPI Timing Characteristics

PTX130W has an exclusive IRQ line used to signal the host a communication request. For SPI communication asserting an IRQ is the only possibility to initialize a transmission from PTX130W to the host.

5 FW/SW Functionality

5.1 NSC Interface

To access PTX130W, an optimized high-level software interface is implemented providing functions for device configuration as well as all data communication. The interface is command- response based.

Data packages between PTX130W and the host are called NSC messages, and the protocol related header bytes are added by the HW-accelerator automatically.

Panthronics provides additional SW-stacks on top of the NSC interface to further ease the integration of the PTX130W into the target application. For PTX130W, besides lower-level APIs like Reader, Tag-type, NDEF etc., a dedicated API for NFC-WLC as standalone application is also provided.

5.2 SW Split Stack

Panthronics provides additional SW-stacks on top of the NSC interface to further ease the integration of the PTX130W into the target application. The SW stacks manage all interactions with the PTX130W on NSC level, by setting up and configuring the device, consolidating status information, handling error messages, and establishing a data channel between the host and the NFC controller.

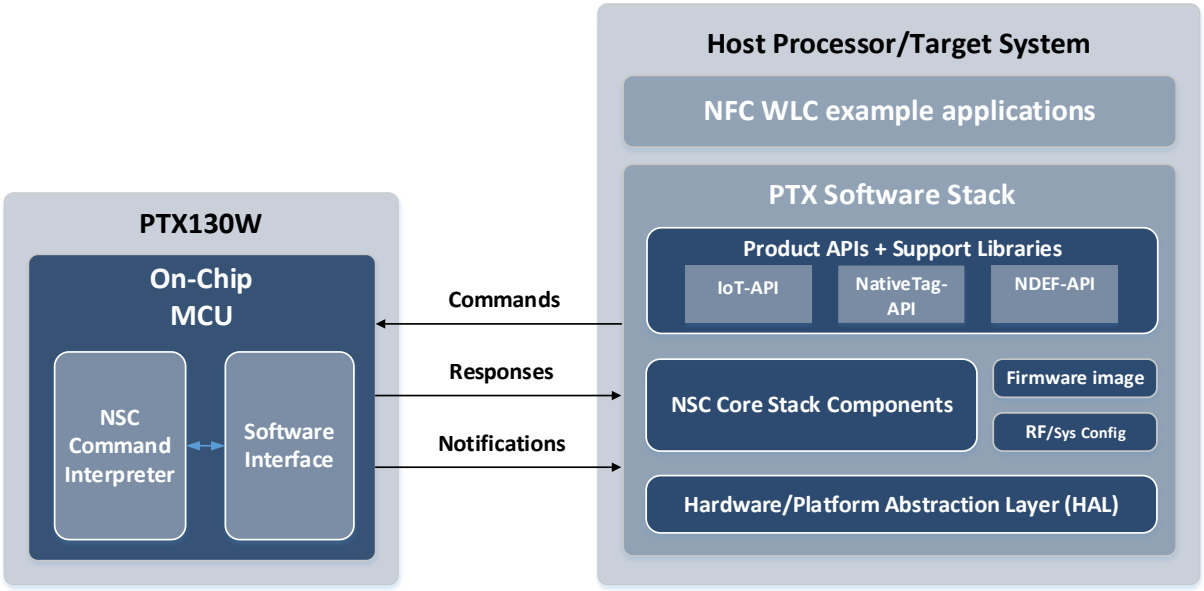


Figure 7: PTX130R SW Stack Integration View

For more details on the PTX130W SW integration, please refer to the integration manual accordingly.

6 Reference Schematic

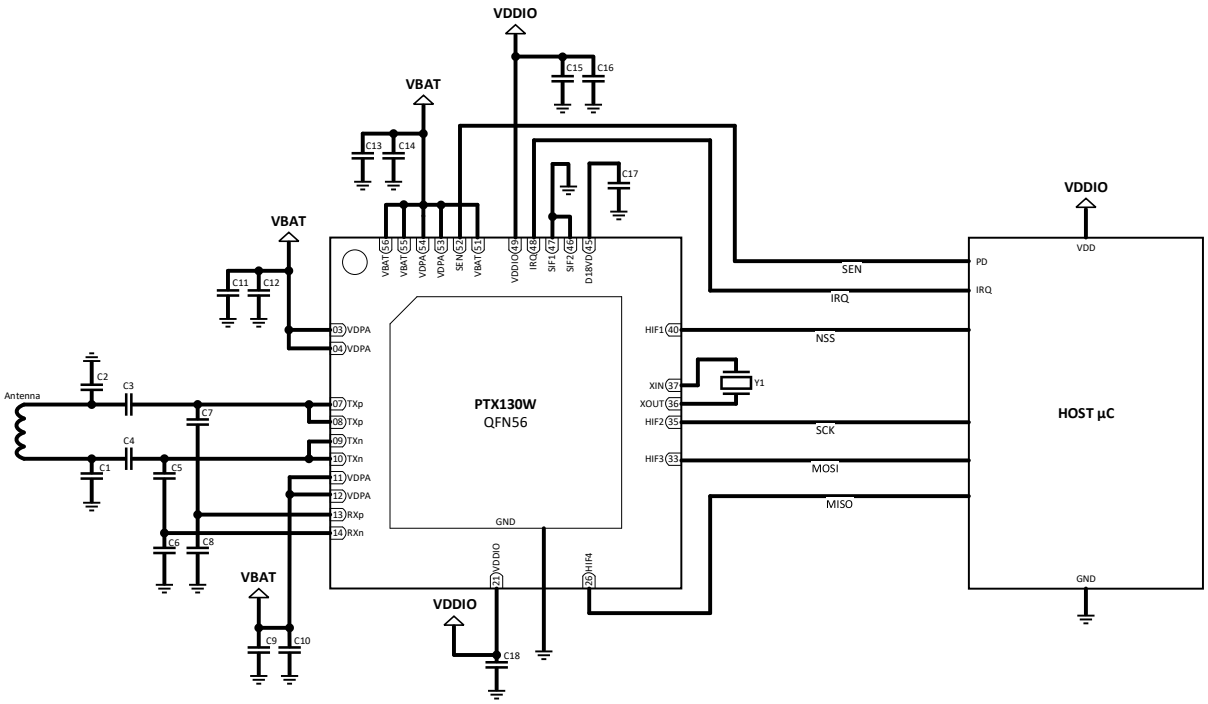


Figure 8: Exemplary Reference Schematic

Designator	Component type	Component value	Description
C1, C2, C3, C4, C5, C6, C7, C8	Ceramic capacitor, C0G type	-	Matching capacitors
C9, C11, C13	Ceramic capacitor, X7R or X5R type	10µF	Note that capacitor shall have at least 2.2µF effective capacitance at applied voltage
C10, C12, C14, C15, C17, C18	Ceramic capacitor, X7R or X5R type	100nF	
C16	Ceramic capacitor, C0G type	10µF	Optional, depending upon VDDIO supply noise/impedance
Y1	Crystal oscillator	27.12MHz	According to Table 7

Table 10: Reference component values

7 Package Information

7.1 Package Marking

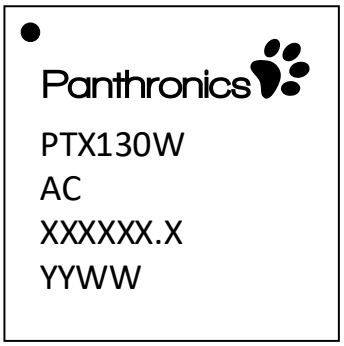
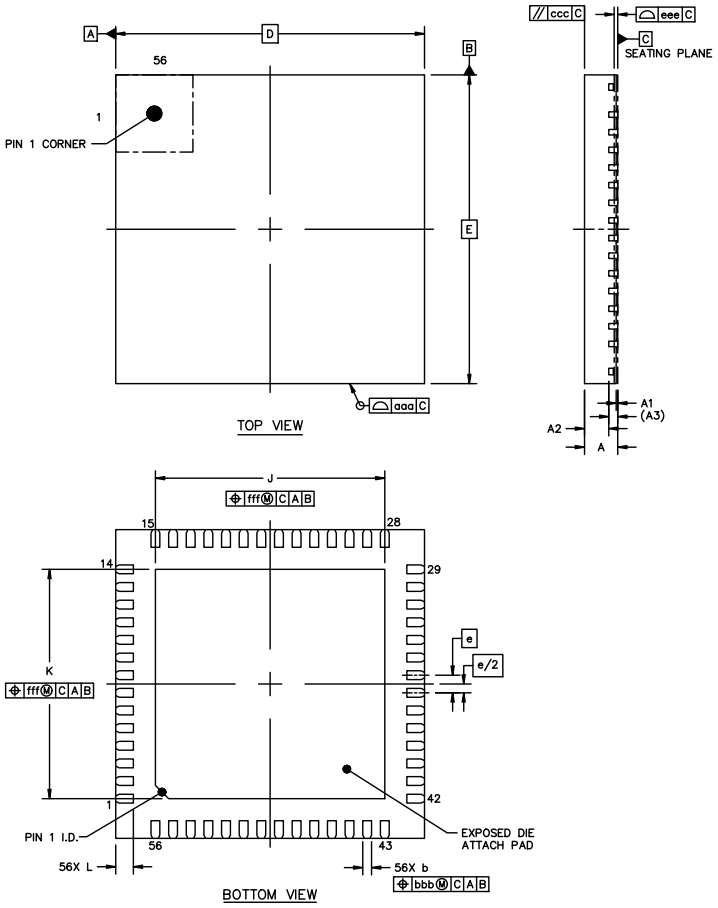


Figure 9: Package marking

Symbol	Description
PTX130W	Device name
AC	Fix characters
XXXXXX.X	Wafer lot number in the following format (i.e., Q22H43.1)
YYWW	Date code (production year and week)

Table 11: Package marking code

7.2 Package Drawing and Dimension



		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.7	0.75	0.8
STAND OFF		A1	0	0.035	0.05
MOLD THICKNESS		A2	---	0.55	---
L/F THICKNESS		A3		0.203 REF	
LEAD WIDTH		b	0.15	0.2	0.25
BODY SIZE	X	D		7 BSC	
	Y	E		7 BSC	
LEAD PITCH		e		0.4 BSC	
EP SIZE	X	J	5.1	5.2	5.3
	Y	K	5.1	5.2	5.3
LEAD LENGTH		L	0.3	0.4	0.5
PACKAGE EDGE TOLERANCE		aaa		0.1	
LEAD OFFSET		bbb		0.1	
MOLD FLATNESS		ccc		0.1	
COPLANARITY		eee		0.08	
EXPOSED PAD OFFSET		fff		0.1	

NOTES
1.0 COPLANARITY APPLIES TO LEADS, CORNER LEADS AND DIE ATTACH PAD.
2.0 TOTAL THICKNESS NOT INCLUDE SAW BURR.

Figure 10: Package HVQFN56 Drawings and Dimensions (Unit: mm)

8 Revision History

Revision	Date	Description
1.0	August 4, 2023	Initial release.
1.1	August 28, 2023	Added information about the limitation regarding multidevice SPI interface support.